MILITARY SPECIFICATION

TEST SET SUBASSEMBLY: 11726303
(Test Socket Assembly)

1. SCOPE

This specification covers a test set subassembly, hereinafter called test socket assembly, used in Test Set, Subassembly MX-8050/TPM-22, part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Process and Inspection of

FSC 4931

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F11726303	Test Socket Assembly

PACKAGING DATA SHEET

11726303	Packaging of Test Socket Assembly

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The test socket assembly shall be manufactured in accordance with Drawing F11726303 and drawings pertaining thereto.

3.2 General specification. - Where applicable, the test socket assembly shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The test socket assembly shall be mounted on the vibration table in normal service position. The test socket assembly shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The test socket assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

3.3.3 Altitude.

3.3.3.1 Operating. - The test socket assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The test socket assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The test socket assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage). - The test socket assembly shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when turned to room ambient.

3.3.5 Humidity-nonoperating. - The test socket assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. - Each conductor shall be continuous electrically as specified on the schematic diagram.

3.4.2 Resistance. - Resistance shall be as given below between the indicated terminals.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.

Terminals	Resistance (ohms)
J5-A through NN and corresponding terminals of J6 (56 pairs of terminals).	470 ± 10%, each pair of terminals
P2-A and J4-B	100 ± 5%
P2-D and J4-A	120 ± 5%
P2-H and J4-A	120 ± 5%
J4-A and J4-D	330 ± 5%
J4-A and J4-H	330 ± 5%
J4-A and J4-L	330 ± 5%
J7-L(+) and J7-N(-)	less than 0.3 (R_{BE} of Q1)
J7-N(+) and J7-L(-)	more than 150 (R_{EB} of Q1)
J7-L(+) and J7-X(-)	less than 0.3 (R_{BC} of Q1)
J7-X(+) and J7-L(-)	more than 150 (R_{CB} of Q1)
J7-R(+) and J7-r(-)	less than 0.3 (R_{BE} of Q2)
J7-r(+) and J7-R(-)	more than 150 (R_{EB} of Q2)
J7-R(+) and J7-T(-)	less than 0.3 (R_{BC} of Q2)
J7-T(+) and J7-R(-)	more than 150 (R_{CB} of Q2)

Source: https://assist.dla.mil -- Downloaded: 20260-22T20:22Z
Check the source to verify that this is the current version before use.

Terminals	Resistance (ohms)
J7-y(+) and J7-w(-)	less than 0.3 (R_{BE} of Q3)
J7-w(+) and J7-y(-)	more than 150 (R_{EB} of Q3)
J7-y(+) and J7-EE(-)	less than 0.3 (R_{BC} of Q3)
J7-EE(+) and J7-y(-)	more than 150 (R_{CB} of Q3)
J7-CC(+) and J7-MM(-)	less than 0.3 (R_{BE} of Q4)
J7-MM(+) and J7-CC(-)	more than 150 (R_{EB} of Q4)
J7-CC(+) and J7-AA(-)	less than 0.3 (R_{BC} of Q4)
J7-AA(+) and J7-CC(-)	more than 150 (R_{CB} of Q4)

3.5 Performance.

3.5.1 Digiswitches. - With 24 ± 1 Vdc applied between J3-K(+) and J3-L(RET), digiswitch lamps shall be illuminated to display the number of the J6 test socket pin to which each digiswitch has been set. The TEST POINT A digiswitch shall serve to connect its selected test socket pin to J8 (test point "A"). The TEST POINT B digiswitch shall serve to connect its selected test socket pin to J10 (test point "B"). The connections so established shall not exceed 0.5 ohms resistance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

3.5.2 Relay K1. - The application of 24 ± 1 Vdc between J3-A(+) and J3-B(RET) shall cause relay K1 to operate, establishing contact between the pairs of terminals shown in columns A and B below. Removal of voltage between J3-A & B shall cause the reverse procedure to occur.

A	B
J4-A	P1-AJ
J4-A	P1-AK
J4-A	P1-AL
J4-B	P1-AB
J4-B	P1-AC
J4-B	P1-AD

Resistance of the circuits so established shall be less than 0.5 ohms.

3.5.3 Relay K2. - The application of 24 ± 1 Vdc between J3-C(+) and J3-D(RET) shall cause relay K2 to operate, breaking contact between the pairs of terminals shown in columns A and B below, and establishing contact between the pairs of terminals shown in columns A and C. Removal of voltage between J3-C&D shall cause the reverse procedure to occur.

A	B	C
J5-y	P1-AY	J2-D
J5-w	P1-AW	J2-A
J5-W	P1-W	J2-T
J5-s	P1-AS	J2-H
J5-u	P1-AU	J2-L
J5-CC	P1-BC	J2-P

Resistance of the circuits so established shall be less than 0.5 ohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

3.5.4 Relay K3. - The application of 24 ± 1 Vdc between J3-E(+) and J3-F(RET) shall cause relay K3 to operate, breaking contact between the pairs of terminals in columns A and B shown below, and establishing contact between the pairs of terminals shown in columns A and C. Removal of voltage between J3-E&F shall cause the reverse procedure to occur.

A	B	C
J5-w	P1-AW	-
J5-y	P1-AY	J2-FF

Resistance of the circuits so established shall be less than 0.5 ohms.

3.5.5 Relay K4. - The application of 24 ± 1 Vdc between J3-H(+) and J3-J(RET) shall cause relay K4 to operate, breaking contact between the pairs of terminals in columns A and B shown below, and establishing contact between the pairs of terminals shown in columns A and C. Removal of voltage between J3-H&J shall cause the reverse procedure to occur.

A	B	C
J5-N	P1-N	J2-W
J5-W	P1-W	J2-Z

Resistance of the circuits so established shall be less than 0.5 ohms.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 20260-22T20:22Z
Check the source to verify that this is the current version before use.

4.1.1 General provisions. - The component and sub-assembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one test socket assembly manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

100% INSPECTION

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Continuity	3.4.1	4.6.2.1
102. Resistance	3.4.2	4.6.2.2
103. Digiswitches	3.5.1	4.6.3.1
104. Relay K1	3.5.2	4.6.3.2
105. Relay K2	3.5.3	4.6.3.3
106. Relay K3	3.5.4	4.6.3.4
107. Relay K4	3.5.5	4.6.3.5

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One test socket assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

TABLE II - SPECIAL SAMPLING

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specifications	3.2	MIL-F-13926/ MIL-C-46357 visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306.	Altitude - nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment. - In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable test as specified in 4.6:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

TEST EQUIPMENT LIST

Item		Description
1.	Power supply	24.0 ± 1 Vdc, capacity 2A.
2.	Kelvin bridge	Capable of measuring resistances from 0.01 ohms to 1.0 ohm, accuracy 1% of reading plus 1% of full scale of scale being used.
3.	Ohmmeter	Capable of measuring resistances from 1.0 ohm to 1 megohm; accuracy 0.5% of reading plus 0.5% of full scale for values up to 1 kilohm; accuracy 2% of reading plus 1% of full scale for values above 1 kilohm.
4.	Test harness	Per Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

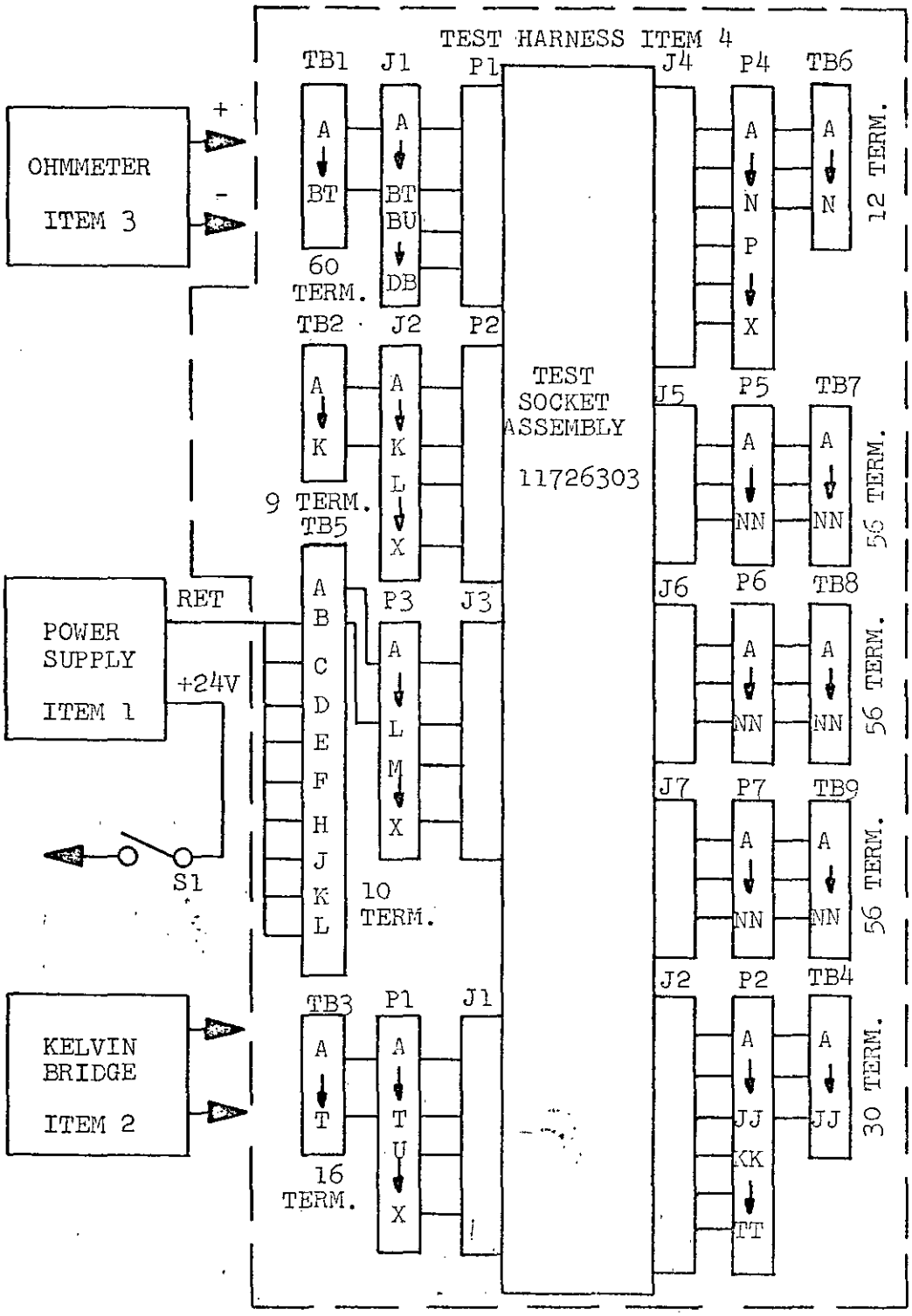


FIGURE 1 TEST SETUP FOR TEST SOCKET ASSEMBLY

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

PARTS LIST FOR TEST HARNESS, ITEM 4

Designation	Part	Description
J1	Connector	Elco 00-8016-090-217-707 or equivalent
J2	Connector	Elco 00-8016-020-217-702 or equivalent
P1	Connector	Elco 00-8016-020-217-709 or equivalent
P2	Connector	Elco 00-8016-038-217-723 or equivalent
P3	Connector	Same as P1
P4	Connector	Same as P1
P5	Connector	Elco 00-8016-056-217-719 or equivalent
P6	Connector	Same as P5
P7	Connector	Same as P5
S1	Switch	SPST, 2A.

Terminal strips, wire, per Figure 1.

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure Criteria of Method 514 of MIL-STD-810 do not apply. After vibration, the test socket assembly shall be tested as given in the sub-paragraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11 ± 1 ms and 15 g's. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the test socket assembly shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating. - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the test socket assembly shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the test socket assembly shall be tested in accordance with 4.6.2 and 4.6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

4.6.1.5 Humidity-nonoperating. - This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical.

4.6.2.1 Continuity. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.6.2.2 Resistance. - Resistance shall be measured with an ohmmeter, item 3 for compliance with 3.4.2.

4.6.3 Performance.

4.6.3.1 Digiswitches. - Adjust power supply, item 1, to deliver 24 ± 1 Vdc and connect between test harness (item 4) terminals TB5-K(+) and L(RET). Close test harness switch S1. Two lamps shall be illuminated in each digiswitch, one in the furthest left section of the digiswitch (the "tens" section), and one in one of the remaining six sections (the "units" sections). Turn digiswitches so that each displays "0" on the furthest left section and "1" on the immediately adjacent section. Connect one lead of Kelvin bridge, item 2, to J8 of test socket assembly, and other lead to test harness TB8-A, and verify that continuity exists. Turn TEST POINT A digiswitch to read "02". Continuity should be broken. Move lead of bridge from TB8-A to TB8-B and verify that continuity exists. Continue procedure, changing digiswitch position by increments of one and moving bridge lead, and verify all 56 positions of TEST POINT A digiswitch. Use same procedure for J10 of test socket assembly and TEST POINT B digiswitch.

4.6.3.2 Relay K1. - Connect power supply, item 1, between test harness terminals TB5-A(+) and TB5-B(RET). With test harness switch S1 closed, use Kelvin bridge, item 2, to verify that continuity exists between points indicated in 3.5.2. Open switch S1 and verify that continuity is broken.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.

4.6.3.3 Relay K2. - Connect power supply, item 1, between test harness terminals TB5-C(+) and TB5-D(RET). Using Kelvin bridge, item 2, and test harness switch S1, verify that operation and release of relay K2 establish and break continuity as indicated in 3.5.3.

4.6.3.4 Relay K3. - Connect power supply, item 1, between test harness terminals TB5-E(+) and TB5-F(RET). Using Kelvin bridge, item 2, and test harness switch S1, verify that operation and release of relay K3 establish and break continuity as indicated in 3.5.4.

4.6.3.5 Relay K4. - Connect power supply, item 1, between test harness terminals TB5-H(+) and TB5-J(RET). Using Kelvin bridge, item 2, and test harness switch S1, verify that operation and release of relay K4 establish and break continuity as indicated in 3.5.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11726303. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The test socket assembly contains receptacles for insertion of printed circuit cards to be tested. The connector pins of these receptacles are connected through two digiswitches to two test points which can be used to monitor the voltages and/or signals at any desired printed circuit card terminals. Some of these terminals which are always used for standard functions (e.g. logic power and return), are connected through relay contacts in the test socket assembly to provide for application of standard power or signals to the printed circuit card under test.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet (see 5.1).

(c) Selection of applicable level of preservation packaging and packing.

(d) Level of protection required.

(e) Applicable stock number.

(f) Block of serial numbers to identify each test socket assembly.

Custodian

Army - MU

Preparing Activity

Army - MU

Project No. 4931-A031

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:22Z
Check the source to verify that this is the current version before use.